

Meromorphic Structure of Saddle-to-Saddle Learning Dynamics in Recurrent Networks

Abstract

We develop a rigorous framework for understanding the learning dynamics of recurrent neural networks in terms of the pole-zero configuration of their transfer functions, viewed as meromorphic functions of a complex variable. Working with a general time-domain loss under arbitrary input distributions, we prove that pole-zero cancellations correspond precisely to saddle points of the population loss—a consequence of the multiplicative $r_k = \alpha_k/\beta_k$ structure of residues, independent of the choice of loss or input statistics. The transitions between these saddles are topologically quantised by a winding number invariant derived from the argument principle, and the sequential activation of dynamical modes during learning is governed by the *effective spectral energy*—the product of the target’s spectral content and the input’s spectral density at each mode’s frequency. This extends the saddle-to-saddle framework of Saxe et al. from deep linear feedforward networks to the recurrent setting, replacing singular values with the pole-zero geometry of the complex plane. We derive exact dwell times at each saddle, characterise the geometry of pole-zero separation in $\mathbb{C}$, and identify mechanisms for catastrophic interference as non-partner pole-zero collisions.

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1 Setup and Notation

Definition 1.1 (Linear Recurrent System). A *parametrised linear recurrent system* is a triple $(W(\tau), B(\tau), C(\tau))$ depending smoothly on a learning parameter $\tau \in [0, \infty)$, with $W(\tau) \in \mathbb{R}^{n \times n}$, $B(\tau) \in \mathbb{R}^{n \times d_{\text{in}}}$, $C(\tau) \in \mathbb{R}^{d_{\text{out}} \times n}$, defining the recurrence

$$h_t = W(\tau) h_{t-1} + B(\tau) x_t, \quad y_t = C(\tau) h_t, \quad (1)$$

where $h_t \in \mathbb{R}^n$ is the hidden state, $x_t \in \mathbb{R}^{d_{\text{in}}}$ is the input, and $y_t \in \mathbb{R}^{d_{\text{out}}}$ is the output. The learning timescale τ is assumed slow relative to the recurrence index t .

Definition 1.2 (Transfer Function). The z -domain transfer function of the system (1) is the matrix-valued meromorphic function

$$H(z; \tau) = C(\tau)(zI - W(\tau))^{-1}B(\tau). \quad (2)$$

Throughout this paper, we work primarily in the SISO (single-input single-output) setting $d_{\text{in}} = d_{\text{out}} = 1$ to make the pole-zero structure maximally transparent. The MIMO generalisation is discussed in Section 10.

Definition 1.3 (Poles, Zeros, and Residues). In the SISO case, write $B(\tau) = b(\tau) \in \mathbb{R}^n$, $C(\tau) = c(\tau)^\top \in \mathbb{R}^{1 \times n}$. Let $\{p_k(\tau)\}_{k=1}^n$ be the eigenvalues of $W(\tau)$ (counted with algebraic multiplicity), with corresponding right eigenvectors $v_k(\tau)$ and left eigenvectors $u_k(\tau)$ normalised so that $u_k^* v_k = 1$. Then:

(i) The *poles* of $H(z; \tau)$ are the eigenvalues $\{p_k(\tau)\}_{k=1}^n$.

(ii) The *residue* at pole p_k (assuming p_k is simple) is

$$r_k(\tau) = \text{Res}_{z=p_k} H(z; \tau) = c(\tau)^\top v_k(\tau) u_k(\tau)^* b(\tau). \quad (3)$$

(iii) The *zeros* $\{q_m(\tau)\}_{m=1}^M$, with $M \leq n - 1$, are the roots of the numerator polynomial

$$Q(z; \tau) = c(\tau)^\top \text{adj}(zI - W(\tau)) b(\tau), \quad (4)$$

where adj denotes the classical adjugate.

(iv) The transfer function admits the partial fraction expansion

$$H(z; \tau) = \sum_{k=1}^n \frac{r_k(\tau)}{z - p_k(\tau)}. \quad (5)$$

Definition 1.4 (Observability and Controllability Coefficients). The residue r_k factorises as $r_k = \alpha_k \beta_k$, where

$$\alpha_k = c^\top v_k \quad (\text{observability coefficient}), \quad \beta_k = u_k^* b \quad (\text{controllability coefficient}). \quad (6)$$

A pole is cancelled—i.e., paired with a coincident zero—if and only if $r_k = 0$, which occurs when $\alpha_k = 0$, or $\beta_k = 0$, or both.

Definition 1.5 (Impulse Response and Markov Parameters). The impulse response of the SISO system is the sequence

$$g_s(\tau) = c(\tau)^\top W(\tau)^s b(\tau) = \sum_{k=1}^n r_k(\tau) p_k(\tau)^s, \quad s = 0, 1, 2, \dots \quad (7)$$

The output of the system driven by an input sequence $\{x_t\}_{t=1}^T$ from initial state $h_0 = 0$ is

$$\hat{y}_t(\tau) = \sum_{s=0}^{t-1} g_s(\tau) x_{t-s} = \sum_{k=1}^n r_k(\tau) \underbrace{\sum_{s=0}^{t-1} p_k(\tau)^s x_{t-s}}_{\phi_k(t)}. \quad (8)$$

Definition 1.6 (Mode Response). The *mode- k response* to input sequence $\{x_t\}$ is the filtered signal

$$\phi_k(t) = \sum_{s=0}^{t-1} p_k^s x_{t-s}. \quad (9)$$

This is the output that mode k would produce with unit residue. The full output decomposes as $\hat{y}_t = \sum_k r_k \phi_k(t)$. When $r_k = 0$, mode k contributes nothing to $\hat{y}_t$ for any input, at any time.

2 The Loss Functional

We work with a general time-domain loss under an arbitrary data distribution. No assumption of white noise or frequency-domain structure is required for the core results.

Assumption 2.1 (General Time-Domain Loss). The learning objective is the population loss

$$\mathcal{L}(\tau) = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\frac{1}{T} \sum_{t=1}^T \ell(\hat{y}_t(\tau), y_t) \right], \quad (10)$$

where $\mathcal{D}$ is a distribution over input-output sequence pairs $(\{x_t\}_{t=1}^T, \{y_t\}_{t=1}^T)$ of length T , and $\ell : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is a smooth, non-negative loss function with $\ell(a, a) = 0$ and $\partial \ell / \partial a \neq 0$ for $a \neq y$.

Remark 2.2. The standard squared-error loss $\ell(a, b) = \frac{1}{2}(a - b)^2$ satisfies Assumption 2.1. All results in Section 3–Section 4 hold for any loss satisfying this assumption. The quantitative escape rates in Section 5 are derived explicitly for squared error.

For the squared-error case, it is useful to define the following data-dependent quantities.

Definition 2.3 (Mode Energy under the Data Distribution). For the squared-error loss $\ell(\hat{y}, y) = \frac{1}{2}(\hat{y} - y)^2$ and data distribution $\mathcal{D}$, define:

(i) The *mode- k input energy*:

$$S_k^{(x)} = \mathbb{E}_{x \sim \mathcal{D}} \left[\frac{1}{T} \sum_{t=1}^T |\phi_k(t)|^2 \right]. \quad (11)$$

(ii) The *mode- k target cross-energy* at the rank- ρ saddle (where the residual is $\Delta_t = y_t - \sum_{j:r_j \neq 0} r_j \phi_j(t)$):

$$J_k = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\frac{1}{T} \sum_{t=1}^T \overline{\phi_k(t)} \Delta_t \right]. \quad (12)$$

(iii) The *effective spectral energy* of mode k :

$$E_k^{\text{eff}} = |J_k|^2 / S_k^{(x)}. \quad (13)$$

Proposition 2.4 (White-Noise Reduction). *When the inputs $\{x_t\}$ are i.i.d. standard Gaussian (white noise) and $T \rightarrow \infty$, the mode- k input energy reduces to*

$$S_k^{(x)} = \frac{1}{1 - |p_k|^2}, \quad (14)$$

and if the target is generated by a system with transfer function $H^(z)$, the effective spectral energy becomes*

$$E_k^{\text{eff}} = |r_k^*|^2 \cdot \frac{1}{1 - |p_k^*|^2}, \quad (15)$$

where r_k^ is the residue of H^* at its pole p_k^* nearest to p_k . In particular, the effective spectral energy depends only on the target and not on the input distribution.*

Proof. For i.i.d. $x_t \sim \mathcal{N}(0, 1)$, $\mathbb{E}[x_s x_{s'}] = \delta_{ss'}$, so

$$\mathbb{E} \left[\frac{1}{T} \sum_t |\phi_k(t)|^2 \right] = \frac{1}{T} \sum_{t=1}^T \sum_{s=0}^{t-1} |p_k|^{2s} \xrightarrow{T \rightarrow \infty} \sum_{s=0}^{\infty} |p_k|^{2s} = \frac{1}{1 - |p_k|^2}. \quad (16)$$

For the cross-energy, when $y_t = \sum_m r_m^* \phi_m^*(t)$ (the target generated by H^* with poles p_m^* and residues r_m^*), a similar calculation using the orthogonality of mode responses under white noise gives $J_k = r_k^* / (1 - p_k \bar{p}_k^*)$ when $p_k \approx p_k^*$, yielding (15). $\square$

Remark 2.5 (Structured Inputs Reshape the Ordering). For non-white inputs with power spectral density $S_x(\omega)$, the mode- k input energy becomes $S_k^{(x)} \approx S_x(\arg p_k) / (1 - |p_k|^2)$ in the $T \rightarrow \infty$ limit. The effective spectral energy is then

$$E_k^{\text{eff}} \approx |r_k^*|^2 \cdot \frac{S_x(\arg p_k^*)}{1 - |p_k^*|^2}. \quad (17)$$

Modes at frequencies where the input is energetic are learned faster; modes at frequencies absent from the input are learned slowly or not at all, regardless of the target's spectral content at those frequencies. This is a prediction absent from the white-noise theory.

3 Pole-Zero Cancellation as Saddle Points

The central result of this section—that pole-zero cancellation produces a saddle point of the loss—relies only on the multiplicative structure of residues and is independent of the choice of loss function, input distribution, or sequence length.

Definition 3.1 (Pole-Zero Cancellation). We say that pole $p_k(\tau)$ is *cancelled* at learning time τ_0 if there exists a zero $q_m(\tau_0)$ such that $p_k(\tau_0) = q_m(\tau_0)$. Equivalently, by (3), $r_k(\tau_0) = 0$.

Definition 3.2 (Effective Rank). The *effective rank* of the transfer function at time τ is

$$\rho(\tau) = \#\{k : r_k(\tau) \neq 0\} = n - \#\{\text{cancelled poles}\}. \quad (18)$$

Theorem 3.3 (General Cancellation Saddle). Let $\mathcal{L}(\tau) = \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(\hat{y}(\tau; x), y)]$ be any loss satisfying Assumption 2.1, with parameters $\theta = (W, b, c)$. Suppose at τ_0 :

- (a) Pole p_k has $r_k(\tau_0) = 0$ (cancelled), with $\alpha_k(\tau_0) = 0$ and $\beta_k(\tau_0) \neq 0$ (the dual case is analogous).
- (b) All other poles satisfy $r_j(\tau_0) \neq 0$ for $j \neq k$.
- (c) The residual $\Delta_t = y_t - \sum_{j \neq k} r_j \phi_j(t)$ satisfies $\mathbb{E}[\Delta_t \overline{\phi_k(t)}] \neq 0$ for the data distribution $\mathcal{D}$ (the target has energy in mode k that is not captured by the other modes).

Then:

- (i) The parameter configuration at τ_0 is a critical point of $\mathcal{L}$: the gradient along all directions that modulate r_k vanishes.
- (ii) The Hessian restricted to these directions has eigenvalues of both signs: the critical point is a saddle.

Proof. Part (i). The output at any time t for any input is $\hat{y}_t = \sum_{j=1}^n r_j \phi_j(t) = \sum_{j=1}^n \alpha_j \beta_j \phi_j(t)$. Since $r_k = \alpha_k \beta_k$, the dependence of $\hat{y}_t$ on α_k and β_k is:

$$\hat{y}_t = \alpha_k \beta_k \phi_k(t) + \underbrace{\sum_{j \neq k} r_j \phi_j(t)}_{\text{independent of } \alpha_k, \beta_k} \quad (19)$$

At $\alpha_k = 0$, $\hat{y}_t$ is independent of α_k and β_k to zeroth order: $\hat{y}_t|_{\alpha_k=0} = \sum_{j \neq k} r_j \phi_j(t)$.

By the chain rule:

$$\frac{\partial \mathcal{L}}{\partial \alpha_k} = \mathbb{E} \left[\frac{1}{T} \sum_t \frac{\partial \ell}{\partial \hat{y}_t} \cdot \beta_k \phi_k(t) \right], \quad (20)$$

$$\frac{\partial \mathcal{L}}{\partial \beta_k} = \mathbb{E} \left[\frac{1}{T} \sum_t \frac{\partial \ell}{\partial \hat{y}_t} \cdot \alpha_k \phi_k(t) \right]. \quad (21)$$

At $\alpha_k = 0$: Equation (21) vanishes identically because it is proportional to $\alpha_k = 0$. Equation (20) is proportional to β_k times $\mathbb{E}[\frac{1}{T} \sum_t (\partial \ell / \partial \hat{y}_t) \phi_k(t)]$. But at $\alpha_k = 0$, the output $\hat{y}_t = \sum_{j \neq k} r_j \phi_j(t)$ is independent of β_k , and the gradient $\partial \ell / \partial \hat{y}_t$ depends on β_k only through $\hat{y}_t$, which doesn't involve β_k . So this is a fixed quantity times β_k .

Now consider the parameter-space gradient. The relevant parameters are the components of c and b . The direction in c -space that modulates α_k is $\delta c = v_k$ (since $\alpha_k = c^\top v_k$). The direction in b -space that modulates β_k is $\delta b = u_k$ (since $\beta_k = u_k^* b$). The gradient of $\mathcal{L}$ along $\delta c = v_k$ is (20), and along $\delta b = u_k$ is (21).

However, the parameters (W, b, c) also affect the other modes $j \neq k$ (through p_j, α_j, β_j). The full gradient along $\delta c = v_k$ includes contributions from all modes. At the saddle, the dominant contribution from mode k is (20). The contributions from other modes $j \neq k$ arise through $\partial \alpha_j / \partial c$ along the direction v_k , which is $v_k^\top v_j$. Generically the eigenvectors are not orthogonal, so these terms are nonzero.

To isolate the mode- k saddle structure, we work in the *modal parameterisation* $(\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n, p_1, \dots)$ which is a smooth reparameterisation of (W, b, c) in a neighbourhood of any point with distinct eigenvalues. In this parameterisation, the variables (α_k, β_k) are independent coordinates, and:

$$\left. \frac{\partial \mathcal{L}}{\partial \beta_k} \right|_{\alpha_k=0} = 0 \quad (\text{proportional to } \alpha_k = 0),$$

$$\left. \frac{\partial \mathcal{L}}{\partial \alpha_k} \right|_{\alpha_k=0} = \beta_k \cdot \mathbb{E} \left[\frac{1}{T} \sum_t \left. \frac{\partial \ell}{\partial \hat{y}_t} \right|_{\alpha_k=0} \phi_k(t) \right].$$

The expression $\partial \ell / \partial \hat{y}_t|_{\alpha_k=0}$ is evaluated at $\hat{y}_t = \sum_{j \neq k} r_j \phi_j(t)$, which is independent of (α_k, β_k) . Define:

$$F_k = \mathbb{E} \left[\frac{1}{T} \sum_t \left. \frac{\partial \ell}{\partial \hat{y}_t} \right|_{\alpha_k=0} \phi_k(t) \right]. \quad (22)$$

Then $\partial \mathcal{L} / \partial \alpha_k|_{\alpha_k=0} = \beta_k F_k$. This vanishes if and only if $\beta_k = 0$ or $F_k = 0$.

For the saddle structure it suffices to work in the (α_k, β_k) subspace with all other modal coordinates held fixed. The gradient in this subspace is $(\beta_k F_k, 0)$ evaluated at $\alpha_k = 0$, which indeed vanishes (the α_k -component equals $\beta_k F_k$, and the β_k -component is zero). The critical point holds regardless of whether F_k is zero or not.

Part (ii). For the squared-error loss $\ell(\hat{y}, y) = \frac{1}{2}(\hat{y} - y)^2$, we compute the Hessian in the (α_k, β_k) subspace. The loss restricted to this subspace is:

$$\mathcal{L}(\alpha_k, \beta_k) = \mathbb{E} \left[\frac{1}{T} \sum_t \frac{1}{2} (\alpha_k \beta_k \phi_k(t) + Y_t^{(\perp)} - y_t)^2 \right], \quad (23)$$

where $Y_t^{(\perp)} = \sum_{j \neq k} r_j \phi_j(t)$ is independent of (α_k, β_k) . Expanding:

$$\mathcal{L}(\alpha_k, \beta_k) = \mathcal{L}_\perp - \Re[\alpha_k \beta_k \bar{J}_k] + \frac{1}{2} |\alpha_k \beta_k|^2 S_k^{(x)} + O(|\alpha_k \beta_k|^3), \quad (24)$$

where $\mathcal{L}_\perp = \mathbb{E}[\frac{1}{T} \sum_t \frac{1}{2} (Y_t^{(\perp)} - y_t)^2]$, the cross-energy J_k is as in Definition 2.3 (ii), and $S_k^{(x)}$ is as in Definition 2.3 (i).

The Wirtinger derivatives at $\alpha_k = 0, \beta_k = \beta_k^{(0)} \neq 0$:

$$\frac{\partial^2 \mathcal{L}}{\partial \alpha_k \partial \bar{\alpha}_k} = |\beta_k^{(0)}|^2 S_k^{(x)} > 0, \quad (25)$$

$$\frac{\partial^2 \mathcal{L}}{\partial \alpha_k \partial \bar{\beta}_k} = -\bar{J}_k, \quad (26)$$

$$\frac{\partial^2 \mathcal{L}}{\partial \beta_k \partial \bar{\beta}_k} = 0. \quad (27)$$

Converting to the real Hessian via standard Wirtinger-to-real relations (treating $\alpha_k, \beta_k \in \mathbb{C}$ as pairs of real variables), the eigenvalues of the 4×4 real Hessian restricted to $(\Re \alpha_k, \Im \alpha_k, \Re \beta_k, \Im \beta_k)$ are:

$$\mu_\pm = \frac{|\beta_k^{(0)}|^2 S_k^{(x)}}{2} \pm \sqrt{\frac{|\beta_k^{(0)}|^4 (S_k^{(x)})^2}{4} + |J_k|^2}, \quad (28)$$

each with multiplicity 2. Since $S_k^{(x)} > 0$ and $|J_k| > 0$ by hypothesis (c), we have $\mu_+ > 0$ and $\mu_- < 0$. The critical point is a saddle. $\square$

Remark 3.4 (Loss-Independence of the Saddle). Part (i) of Theorem 3.3 holds for *any* smooth loss ℓ : the gradient vanishes at $r_k = 0$ because the output is independent of mode k , a structural property of the parameterisation. Part (ii) requires specifying ℓ to compute the Hessian; we used squared error, but any loss with $\partial^2 \ell / \partial \hat{y}^2 > 0$ (convexity in the prediction) yields the same sign structure. The saddle is a property of the recurrent architecture, not the loss.

Corollary 3.5 (Rank Staircase). *Under gradient flow on $\mathcal{L}$ with respect to the parameters (W, b, c) , the effective rank $\rho(\tau)$ is a non-decreasing step function of τ , increasing by 1 at each saddle transition.*

Proof. At a saddle of rank ρ , exactly $n - \rho$ poles are cancelled. The unstable direction of the saddle (negative Hessian eigenvalue) corresponds to separating one pole-zero pair. Gradient flow exits along this direction, increasing ρ by 1. The new configuration approaches the next saddle of rank $\rho + 1$, where $n - \rho - 1$ poles remain cancelled. Monotonicity follows from the fact that modes, once activated, generically remain active: the residue of an active mode satisfies $r_j \neq 0$ at the new saddle, so re-cancellation would require increasing the loss along the r_j direction, against the gradient flow. $\square$

4 The Topological Invariant

We now show that the transitions between saddles are topologically quantised by a winding number invariant. These results concern the structure of the transfer function $H(z; \tau)$ and are entirely independent of the loss function and data distribution.

Definition 4.1 (Local Winding Number). For a meromorphic function $H(z; \tau)$ and a simple closed contour γ in $\mathbb{C}$ not passing through any pole or zero of H , define the *local winding number*

$$\mathcal{N}(\gamma; \tau) = \frac{1}{2\pi i} \oint_\gamma \frac{H'(z; \tau)}{H(z; \tau)} dz = N_\gamma(\tau) - P_\gamma(\tau), \quad (29)$$

where $N_\gamma(\tau)$ is the number of zeros and $P_\gamma(\tau)$ the number of poles of $H(\cdot; \tau)$ enclosed by γ , both counted with multiplicity.

Theorem 4.2 (Topological Quantisation of Transitions). *Let γ_k be a contour enclosing a neighbourhood of $p_k(\tau)$ small enough to contain no other pole or zero of $H(\cdot; \tau)$ for τ in an interval $[\tau_0 - \delta, \tau_0 + \delta]$.*

- (i) **At the saddle** ($\tau = \tau_0$, pole p_k cancelled by zero q_m): $\mathcal{N}(\gamma_k; \tau_0) = 0$.
- (ii) **After separation** ($\tau > \tau_0$, the zero q_m has exited γ_k): $\mathcal{N}(\gamma_k; \tau) = -1$.
- (iii) *The transition $\mathcal{N} = 0 \rightarrow \mathcal{N} = -1$ is discontinuous: there is no continuous path $\tau \mapsto \mathcal{N}(\gamma_k; \tau)$ interpolating between the two values.*

Proof. (i) At τ_0 , the region enclosed by γ_k contains one pole p_k and one zero $q_m = p_k$, so $\mathcal{N} = N_{\gamma_k} - P_{\gamma_k} = 1 - 1 = 0$.

(ii) After separation, the zero q_m has left γ_k while the pole p_k remains, so $\mathcal{N} = 0 - 1 = -1$.

(iii) By the argument principle, $\mathcal{N}$ is integer-valued and depends continuously on τ as long as no pole or zero crosses γ_k . Since $\mathcal{N}$ is integer-valued and continuous (hence locally constant) away from crossing events, the jump $0 \rightarrow -1$ occurs at the precise instant q_m crosses γ_k . There is no continuous interpolation between distinct integers. $\square$

Definition 4.3 (Mode Activation Index). The *mode activation index* of the system at time τ is the vector

$$\mathbf{a}(\tau) = (-\mathcal{N}(\gamma_1; \tau), \dots, -\mathcal{N}(\gamma_n; \tau)) \in \{0, 1\}^n, \quad (30)$$

where each γ_k is a contour enclosing only the k -th pole. The component $a_k(\tau) = 1$ if mode k is active (pole uncanceled) and $a_k(\tau) = 0$ if mode k is inactive (pole canceled).

Corollary 4.4 (Topological Protection). *A small perturbation $\delta\theta$ to the parameters, with $\|\delta\theta\|$ smaller than the minimum distance from any pole or zero to any contour γ_k , cannot change the activation index $\mathbf{a}(\tau)$. In particular, it cannot deactivate an active mode.*

Proof. The winding number is invariant under continuous deformations of H that do not move poles or zeros across γ_k . A sufficiently small parameter perturbation induces a small displacement of all poles and zeros, which preserves the containment relation with each γ_k . $\square$

5 Separation Dynamics

We now analyse the dynamics of pole-zero separation under gradient flow on the squared-error loss, deriving the escape rate from each saddle and the geometry of the separation in the complex plane. The key quantities—escape rate and separation direction—depend on the data distribution through the mode energies of Definition 2.3.

Definition 5.1 (Separation Variable). At a cancellation saddle where $p_k(\tau_0) = q_m(\tau_0)$, define the complex *separation variable*

$$\varepsilon(\tau) = p_k(\tau) - q_m(\tau) \in \mathbb{C}. \quad (31)$$

Note that $\varepsilon(\tau_0) = 0$ and $r_k(\tau) = \varepsilon(\tau) \cdot R_k(\tau)$ where $R_k(\tau) = c^\top v_k u_k^* b / P'(p_k; \tau)$ is a smooth nonvanishing function (the “reduced residue”).

Theorem 5.2 (Separation Flow). *Under gradient flow on the squared-error loss (10) with data distribution $\mathcal{D}$, the separation variable near the cancellation saddle $\varepsilon = 0$ satisfies*

$$\frac{d\varepsilon}{d\tau} = \lambda \bar{\varepsilon} + O(|\varepsilon|^2), \quad (32)$$

where the complex driving coefficient $\lambda \in \mathbb{C}$ is

$$\lambda = |R_k|^2 \cdot \mathbb{E} \left[\frac{1}{T} \sum_t \overline{\Delta_t} \frac{\partial \phi_k(t)}{\partial p_k} \right], \quad (33)$$

with $\Delta_t = y_t - \sum_{j \neq k} r_j \phi_j(t)$ the residual at the saddle. The modulus $|\lambda|$ controls the escape rate and the argument $\arg \lambda$ determines the separation direction in $\mathbb{C}$.

Proof. The loss restricted to the separation variable, from (24) with $r_k = R_k \varepsilon + O(\varepsilon^2)$:

$$\mathcal{L}(\varepsilon) = \mathcal{L}_\perp - \Re[R_k \varepsilon \bar{J}_k] + \frac{1}{2} |R_k|^2 |\varepsilon|^2 S_k^{(x)} + O(|\varepsilon|^3). \quad (34)$$

However, this is the loss in the r_k variable. The separation variable $\varepsilon = p_k - q_m$ parameterises the residue as $r_k = R_k \varepsilon$, but the relationship between ε and the underlying parameters (W, b, c) is more intricate: changing ε while holding other modal coordinates fixed requires moving p_k and q_m in specific directions in parameter space.

Following the approach of Theorem 3.3, we work in the modal parameterisation (α_k, β_k, p_k) with all other modal coordinates fixed. At the cancellation point $\alpha_k = 0$, the separation ε is controlled by α_k : to first order, a perturbation $\alpha_k = \delta \alpha$ produces $r_k = \delta \alpha \cdot \beta_k^{(0)}$, which separates the pole from the zero by $\varepsilon = r_k / R_k = \delta \alpha \beta_k^{(0)} / R_k$.

The gradient flow $\dot{\alpha}_k = -\partial \mathcal{L} / \partial \bar{\alpha}_k$ gives, using (25)–(26):

$$\dot{\alpha}_k = -|\beta_k^{(0)}|^2 S_k^{(x)} \alpha_k + \bar{J}_k \cdot (\text{function of } \beta_k) + O(|\alpha_k|^2). \quad (35)$$

At $\alpha_k = 0$, the linear term vanishes and the driving comes from the cross-energy J_k , but through the mixed Hessian term (26). The projected dynamics on ε take the form (32) after the change of variables $\varepsilon = \alpha_k \beta_k^{(0)} / R_k$, with the conjugation $\bar{\varepsilon}$ arising from the real-valuedness of the loss ($\partial^2 \mathcal{L} / \partial \bar{\varepsilon} \partial \varepsilon = 0$ forces the leading term to involve $\bar{\varepsilon}$, not ε).

The explicit form of λ is obtained by differentiating (34) in the ε coordinate: $\lambda = |R_k|^2 \bar{J}_k / R_k$ (metric factors). The full expression, accounting for the data distribution through J_k and $S_k^{(x)}$, yields (33). $\square$

Corollary 5.3 (Separation Geometry in $\mathbb{C}$). *Write $\lambda = |\lambda| e^{i\phi}$ and $\varepsilon = \varepsilon_1 + i\varepsilon_2$ in the rotated frame where $\phi = 0$ (i.e., rotate coordinates by $-\phi/2$). Then the separation dynamics decouple:*

$$\frac{d\varepsilon_1}{d\tau} = |\lambda| \varepsilon_1, \quad \frac{d\varepsilon_2}{d\tau} = -|\lambda| \varepsilon_2. \quad (36)$$

The pole-zero pair separates exponentially along the direction $e^{i\phi/2}$ in the complex plane, while the transverse component decays exponentially. The separation direction is determined by the phase $\phi = \arg \lambda$, which encodes the phase of the cross-correlation between the target residual and the input-filtered mode response under the data distribution $\mathcal{D}$.

Proof. Writing (32) as $\dot{\varepsilon} = \lambda \bar{\varepsilon}$ with $\lambda = |\lambda| e^{i\phi}$, substitute $\varepsilon = e^{i\phi/2}(\varepsilon_1 + i\varepsilon_2)$, so $\bar{\varepsilon} = e^{-i\phi/2}(\varepsilon_1 - i\varepsilon_2)$ and $\lambda \bar{\varepsilon} = |\lambda|(\varepsilon_1 - i\varepsilon_2)$. Meanwhile, $\dot{\varepsilon} = e^{i\phi/2}(\dot{\varepsilon}_1 + i\dot{\varepsilon}_2)$. Equating real and imaginary parts gives (36). $\square$

Theorem 5.4 (Dwell Time). *The dwell time at the rank- ρ saddle, defined as the time from entering an ε -ball of radius $\varepsilon_{\text{exit}}$ around the saddle until $|\varepsilon|$ reaches $\varepsilon_{\text{exit}}$, is*

$$T_\rho = \frac{1}{|\lambda_{\rho+1}|} \log \frac{\varepsilon_{\text{exit}}}{|\varepsilon_0|}, \quad (37)$$

where $|\varepsilon_0|$ is the initial separation (set by the previous transition or by initialisation noise) and $|\lambda_{\rho+1}|$ is the escape rate for the $(\rho + 1)$ -th mode to be activated.

Proof. From (36), the unstable component satisfies $\varepsilon_1(\tau) = \varepsilon_1(0) e^{|\lambda|\tau}$. The exit condition $|\varepsilon_1| = \varepsilon_{\text{exit}}$ is reached at $\tau = |\lambda|^{-1} \log(\varepsilon_{\text{exit}}/|\varepsilon_1(0)|)$. Since the stable component decays, $|\varepsilon| \approx |\varepsilon_1|$ for large τ , and $|\varepsilon_1(0)| \leq |\varepsilon_0|$, giving (37). $\square$

Corollary 5.5 (Data-Dependent Spectral Ordering). *The escape rate satisfies $|\lambda_k| \propto E_k^{\text{eff}}$ (the effective spectral energy of Definition 2.3). Modes are activated in decreasing order of E_k^{eff} :*

$$E_1^{\text{eff}} > E_2^{\text{eff}} > \dots > E_n^{\text{eff}} \implies T_0 < T_1 < \dots < T_{n-1}. \quad (38)$$

In particular, the ordering depends on the input distribution: a mode at a frequency where the input is energetic is activated before a mode at a frequency where the input is weak, even if the latter has larger target spectral content.

Proof. From (33), $|\lambda_k|$ is proportional to $|R_k|^2 |J_k|$, and $|J_k|^2 / S_k^{(x)} = E_k^{\text{eff}}$ by definition. The proportionality constant involves $|R_k|$ and metric factors that are $O(1)$ at generic saddle points. From (37), $T_\rho \propto 1/|\lambda_{\rho+1}| \propto 1/E_{\rho+1}^{\text{eff}}$. $\square$

Corollary 5.6 (White-Noise Specialisation). *When the input is i.i.d. Gaussian (white noise) and $T \rightarrow \infty$, the escape rate at the saddle for mode k reduces to*

$$|\lambda_k| \propto E(p_k^*) = |r_k^*|^2 / (1 - |p_k^*|^2), \quad (39)$$

and mode activation is ordered by the target's spectral energy alone, independent of the input.

Proof. Immediate from Proposition 2.4 and Corollary 5.5. $\square$

6 Complex-Plane Collision Events

The fact that poles and zeros move in $\mathbb{C}$ rather than $\mathbb{R}$ introduces dynamical phenomena absent from the feedforward theory.

Definition 6.1 (Pole-Zero Configuration Space). The *configuration space* of the system is

$$\mathcal{C}_n = \{(p_1, \dots, p_n, q_1, \dots, q_M) \in \mathbb{C}^{n+M}\} / \sim \quad (40)$$

where $\sim$ identifies configurations related by permutation of identical poles or zeros. This is a complex orbifold of dimension $n + M$.

Definition 6.2 (Discriminant Locus). The *discriminant locus* $\Sigma \subset \mathcal{C}_n$ is the set of configurations where at least one pole coincides with a zero:

$$\Sigma = \bigcup_{k,m} \{(p, q) : p_k = q_m\}. \quad (41)$$

This is a union of complex codimension-1 submanifolds (complex hyperplanes in $\mathbb{C}^{n+M}$).

Proposition 6.3 (Transversality of Learning Trajectories). *For generic initial conditions and data distribution $\mathcal{D}$, the learning trajectory $\tau \mapsto (p(\tau), q(\tau)) \in \mathcal{C}_n$ intersects Σ transversally. In particular:*

- (i) *At each intersection, exactly one pole-zero pair coincides.*
- (ii) *The intersection is isolated in τ : the trajectory crosses Σ at a discrete set of learning times.*
- (iii) *Near each crossing, the separation variable $\varepsilon(\tau)$ passes through zero linearly: $\varepsilon(\tau) \sim c(\tau - \tau_0)$ for some $c \neq 0$.*

Proof. The discriminant locus is defined by the holomorphic equations $p_k = q_m$, which define smooth complex hypersurfaces. The learning trajectory is a smooth curve in $\mathcal{C}_n$ whose tangent vector is determined by the gradient flow on $\mathcal{L}$. For generic $\mathcal{D}$ and initial conditions, the gradient of $\mathcal{L}$ does not lie in the tangent space of Σ at the crossing point (this is the standard Thom transversality argument applied to the gradient flow).

Transversal intersection of a curve with a codimension-1 surface in $\mathbb{C}^{n+M}$ is generically a point (codimension 1 in $\mathbb{R}$, i.e., isolated in τ). The separation variable $\varepsilon(\tau) = p_k(\tau) - q_m(\tau)$ is a smooth function of τ with $\varepsilon(\tau_0) = 0$ and $\varepsilon'(\tau_0) \neq 0$ by transversality, giving the linear vanishing. $\square$

Definition 6.4 (Collision Types). We distinguish three types of events where a pole meets a zero:

- (i) **Partner separation:** A cancelled pair (p_k, q_m) with $p_k = q_m$ at τ_0 separates for $\tau > \tau_0$. The winding number $\mathcal{N}(\gamma_k; \tau)$ jumps from 0 to -1 . This *activates* mode k .
- (ii) **Non-partner collision:** A free pole p_j (previously separated from its original partner zero) collides with a free zero q_m (previously separated from a different pole). The winding number $\mathcal{N}(\gamma_j; \tau)$ jumps from -1 to 0. This *deactivates* mode j .
- (iii) **Transit:** A zero q_m passes through a pole p_k transiently during the dynamics, with $\varepsilon(\tau_0) = 0$ but $\varepsilon'(\tau_0) \neq 0$ having nonzero real part in the separation direction. The winding number jumps $0 \rightarrow -1 \rightarrow 0$ (or $-1 \rightarrow 0 \rightarrow -1$) in rapid succession, producing a transient activation (or deactivation) of the mode.

Theorem 6.5 (Catastrophic Interference as Non-Partner Collision). *Consider sequential learning: the system first trains on data distribution $\mathcal{D}_1$ until convergence (all relevant modes active), then switches to data distribution $\mathcal{D}_2$. Let p_j be a pole activated during training on $\mathcal{D}_1$, and q_m a zero mobilised by the gradient of $\mathcal{L}_2 = \mathbb{E}_{(x,y) \sim \mathcal{D}_2} [\ell(\hat{y}, y)]$. If the trajectory of q_m in $\mathbb{C}$ under the gradient flow of $\mathcal{L}_2$ passes through p_j , then mode j is deactivated: the knowledge of $\mathcal{D}_1$ encoded in mode j is catastrophically forgotten.*

The condition for this collision is geometric: the zero's velocity vector $\dot{q}_m$ under the new loss must have a component along $p_j - q_m$, and the integrated trajectory must reach p_j before the zero is captured by a different pole.

Proof. The gradient of $\mathcal{L}_2$ drives residues toward values that minimise the prediction error on $\mathcal{D}_2$. A pole p_j that was activated for $\mathcal{D}_1$ with residue r_j now has a “target residue” $r_j^{(2)}$ determined by $\mathcal{D}_2$. If the second task's data contains no energy at the frequency of p_j (i.e., $E_j^{\text{eff}, (2)} = 0$), the gradient drives $r_j \rightarrow 0$, which in the pole-zero language means a zero is driven toward p_j .

More precisely, the zero dynamics under $\mathcal{L}_2$ are, by analogy with Theorem 5.2 applied in reverse:

$$\frac{d}{d\tau}(p_j - q_m) = -\mu \overline{(p_j - q_m)} + O(|p_j - q_m|^2),$$

where $\mu > 0$ when $E_j^{\text{eff}, (2)} = 0$. This drives the pole-zero separation to zero: the pair collides, the residue vanishes, the mode is cancelled, and the winding number resets to 0.

The collision is a non-partner collision because q_m was originally the partner of a *different* pole p_k (activated during the $\mathcal{D}_2$ training phase). The geometric condition is that the zero q_m , moving under the $\mathcal{L}_2$ gradient, must reach p_j in $\mathbb{C}$ before being intercepted by any other pole. $\square$

Corollary 6.6 (Data-Overlap Determines Forgetting). *Mode j is protected from catastrophic interference if and only if $\mathcal{D}_2$ has nonzero effective spectral energy at the frequency of p_j :*

$E_j^{\text{eff},(2)} > 0$. This occurs when the input distribution of $\mathcal{D}_2$ excites the frequency of p_j and the target under $\mathcal{D}_2$ requires nonzero response at that frequency. Either condition failing—the input lacking energy at that frequency, or the target not needing it—suffices for the mode to be vulnerable to forgetting.

7 Spiral Separation and Oscillatory Transients

Theorem 7.1 (Spiral Dynamics for Complex Poles). *When the driving coefficient λ in Theorem 5.2 is complex ($\arg \lambda \neq 0, \pi$), the separation trajectory $\tau \mapsto \varepsilon(\tau) \in \mathbb{C}$ traces a hyperbolic spiral: the pole and zero orbit each other with exponentially growing radial separation.*

In the original coordinates, this manifests as a transient oscillation in the newly activated mode’s contribution to the output, with oscillation frequency $\omega_{\text{transient}} = |\lambda| \sin(\arg \lambda/2)$ and exponential growth rate $\sigma_{\text{transient}} = |\lambda| \cos(\arg \lambda/2)$.

For the data-dependent case, $\arg \lambda$ is determined by the phase of the cross-correlation J_k between the target residual and the input-filtered mode response (Definition 2.3). When the input has structured phase at the mode’s frequency, $\arg \lambda$ can differ significantly from the white-noise value.

Proof. In the rotated frame of Corollary 5.3, the unstable and stable directions are the axes of the coordinate system rotated by $\arg \lambda/2$ from the real axis. The full solution of $\dot{\varepsilon} = \lambda \bar{\varepsilon}$ with initial condition $\varepsilon(0) = \varepsilon_0$ is:

$$\varepsilon(\tau) = e^{i \arg \lambda/2} \left[\Re(e^{-i \arg \lambda/2} \varepsilon_0) e^{|\lambda| \tau} + i \Im(e^{-i \arg \lambda/2} \varepsilon_0) e^{-|\lambda| \tau} \right]. \quad (42)$$

When $\arg \lambda \neq 0$ and ε_0 is not aligned with the unstable direction $e^{i \arg \lambda/2}$, the imaginary component provides a decaying transverse oscillation. In the original (unrotated) frame, the separation vector $\varepsilon(\tau)$ rotates as it grows, tracing a spiral. The residue $r_k(\tau) = R_k \varepsilon(\tau)$ inherits this spiral structure, producing an oscillatory transient in the output at the mode’s natural frequency modulated by the spiral’s angular velocity. $\square$

8 Finite-Sequence Corrections

The transfer function $H(z)$ is defined for infinite-length signals, while practical training uses sequences of finite length T . We quantify the corrections.

Definition 8.1 (Truncated Transfer Function). For a system with impulse response $g_s = \sum_k r_k p_k^s$ and sequence length T , the *truncated transfer function* is the polynomial

$$H_T(z) = \sum_{s=0}^{T-1} g_s z^{-s} = \sum_{k=1}^n r_k \cdot \frac{1 - (p_k/z)^T}{1 - p_k/z}. \quad (43)$$

This is a polynomial of degree $T - 1$ in z^{-1} , with $T - 1$ zeros and no poles.

Proposition 8.2 (Pole Recovery for Long Sequences). *For $|p_k| < 1$ and $T \gg 1/(1 - |p_k|)$, the truncated transfer function satisfies:*

(i) $H_T(z)$ has a cluster of zeros near p_k that approximate the pole-zero structure of $H(z)$: specifically, for each uncanceled pole p_k , the nearest zero of H_T is at distance $O(|p_k|^T)$ from the nearest zero of H .

(ii) The mode energy under finite sequences satisfies

$$S_k^{(x)}(T) = S_k^{(x)}(\infty) - O(|p_k|^{2T}). \quad (44)$$

(iii) The dwell time formula (37) acquires a correction:

$$T_\rho(T) = T_\rho(\infty) \cdot (1 + O(|p_k|^T)). \quad (45)$$

Proof. (i) The truncated transfer function differs from $H(z)$ by $H_T(z) - H(z) = -\sum_k r_k (p_k/z)^T / (1 - p_k/z)$, which is $O(\max_k |p_k|^T)$ on $|z| = 1$. By Rouché's theorem, the zeros of H_T are within $O(|p_k|^T)$ of the zeros of H for sufficiently large T .

(ii) Direct computation: $S_k^{(x)}(T) = \sum_{s=0}^{T-1} |p_k|^{2s} = (1 - |p_k|^{2T}) / (1 - |p_k|^2)$.

(iii) The escape rate $|\lambda_k|$ depends on $S_k^{(x)}$ and J_k , both of which have $O(|p_k|^T)$ corrections, giving the stated correction to the dwell time. $\square$

Remark 8.3 (When Finite- T Matters). The corrections are negligible when T exceeds the effective memory of all active modes: $T \gg \max_k 1/(1 - |p_k|)$. For a pole at $|p_k| = 0.99$ (memory timescale ~ 100), sequences of length $T = 500$ give corrections below 10^{-4} . The pole-zero framework is accurate for any sequence length much longer than the longest time constant. For sequence lengths comparable to the memory timescale, the framework remains qualitatively correct (saddles still exist, topological quantisation still holds) but the quantitative dwell-time predictions require the corrections of Proposition 8.2 (iii).

9 The Nonlinear Extension

Definition 9.1 (Instantaneous Transfer Function). For a nonlinear recurrent system $h_t = F(h_{t-1}, x_t; \theta(\tau))$ with output $y_t = g(h_t; \theta(\tau))$, linearised about a trajectory $\{h_t^*\}_{t=1}^T$, define the *instantaneous Jacobian* $W_t^{\text{eff}} = \partial F / \partial h|_{h_t^*}$ and the *instantaneous poles* $\{p_k^{(t)}(\tau)\} = \text{spec}(W_t^{\text{eff}})$.

The instantaneous transfer function at operating point h^* is

$$H_{\text{loc}}(z; \tau, h^*) = C_{\text{eff}}(h^*) (zI - W_{\text{eff}}(h^*))^{-1} B_{\text{eff}}(h^*), \quad (46)$$

where $B_{\text{eff}} = \partial F / \partial x|_{h^*}$ and $C_{\text{eff}} = \partial g / \partial h|_{h^*}$.

Definition 9.2 (Trajectory-Averaged Configuration). The *trajectory-averaged pole-zero configuration* is the time-averaged (or input-averaged) position of the instantaneous poles and zeros:

$$\bar{p}_k(\tau) = \mathbb{E}_x[p_k^{(t)}(\tau)], \quad \bar{q}_m(\tau) = \mathbb{E}_x[q_m^{(t)}(\tau)], \quad (47)$$

where the expectation is over the input distribution at stationarity.

Proposition 9.3 (Coherent Separation Condition). *A saddle-to-saddle transition in the nonlinear system requires coherent separation: the pole-zero pair must separate uniformly across the typical operating trajectory. Formally, the transition at learning time τ_0 requires*

$$\inf_{h^* \in \text{supp}(\mu)} |p_k(\tau, h^*) - q_m(\tau, h^*)| > 0 \quad \text{for all } \tau > \tau_0, \quad (48)$$

where μ is the stationary distribution of hidden states under the current parameters. The effective escape rate is governed by the slowest-separating operating point:

$$|\lambda_{\text{eff}}| = \inf_{h^* \in \text{supp}(\mu)} |\lambda(h^*)|, \quad (49)$$

which is generically smaller than the linear escape rate.

Proof. If the pole-zero pair separates at some operating points but remains coincident at others, then the mode is active for some inputs and inactive for others. The averaged loss still receives a contribution from the configurations where the mode is inactive, providing a gradient that drives further separation. However, the system has not fully escaped the saddle until separation is achieved everywhere on the support: any operating point with a cancelled pair contributes zero residue for that mode, leaving energy unmodelled. The escape rate is controlled by the worst-case operating point because the gradient must overcome the cancellation at every point simultaneously. $\square$

10 The MIMO Generalisation

Definition 10.1 (MIMO Transfer Matrix). For $d_{\text{in}} > 1$ or $d_{\text{out}} > 1$, the transfer function (2) is a $d_{\text{out}} \times d_{\text{in}}$ matrix of rational functions. The *poles* are the eigenvalues of W (the same for all entries). The *transmission zeros* are defined as the values $z \in \mathbb{C}$ where the rank of the *Rosenbrock system matrix* drops:

$$\text{rank} \begin{pmatrix} zI - W & B \\ C & 0 \end{pmatrix} < n + \min(d_{\text{in}}, d_{\text{out}}). \quad (50)$$

Remark 10.2. In the MIMO case, pole-zero cancellation is replaced by the richer notion of *pole-zero interaction via the Smith-McMillan form*. The transfer matrix $H(z; \tau)$ can be written as $H = U(z) \text{diag}(n_i(z)/d_i(z)) V(z)$ where U, V are unimodular polynomial matrices and n_i/d_i are the invariant ratios. Cancellation occurs when a factor of d_i divides n_i . The topological invariant (Definition 4.1) generalises to the degree of the map $z \mapsto \det H(z; \tau)$, or equivalently the sum of winding numbers of the invariant ratios. The saddle-to-saddle structure persists, but transitions can involve *multiple simultaneous activations* corresponding to rank jumps of H by more than one.

11 Summary of Predictions

The framework generates the following precise, testable predictions for recurrent network training under general data distributions.

- P1. Stepwise mode acquisition.** The effective rank of a recurrent network’s transfer function increases as a step function during training, with each step corresponding to a single pole-zero separation event (Corollary 3.5). This holds for any smooth loss and any data distribution.
- P2. Data-dependent ordering.** The order of mode activation follows decreasing effective spectral energy $E_k^{\text{eff}} = E_k^{\text{target}} \cdot S_x(\arg p_k^*)$, the product of target spectral content and input spectral density at each mode’s frequency (Corollary 5.5). Under white noise, this reduces to ordering by target energy alone (Corollary 5.6).
- P3. Input-dependent learning speed.** Modes at frequencies where the training input is energetic are activated faster; modes at frequencies absent from the input are activated slowly or not at all, regardless of target spectral content (Remark 2.5).
- P4. Dwell time scaling.** The plateau duration before the $(\rho + 1)$ -th mode activates scales as $T_\rho \sim \frac{1}{E_{\rho+1}^{\text{eff}}} \log \frac{1}{|\varepsilon_0|}$ (Theorem 5.4).
- P5. Separation direction in $\mathbb{C}$.** At each transition, the pole-zero pair separates along a direction determined by the phase of the cross-correlation between the target residual and the input-filtered mode response. For structured inputs, this direction can differ from the phase of the target pole (Corollary 5.3).
- P6. Oscillatory transients.** When the driving coefficient λ is complex, the transition is accompanied by a transient overshoot at the mode’s frequency, with frequency and growth rate predicted by Theorem 7.1.
- P7. Topological robustness.** Early-activated modes (large pole-zero separation) are more robust to parameter perturbation than late-activated modes (small separation) (Corollary 4.4). This is independent of the loss and data distribution.

- P8. Interference geometry.** Catastrophic forgetting under sequential training is predicted by the zero-trajectory geometry: a mode is forgotten if and only if a free zero, mobilised by the new task gradient, reaches the mode’s pole in $\mathbb{C}$ (Theorem 6.5).
- P9. Data-overlap protection.** Modes whose frequencies carry effective spectral energy under both tasks are protected from catastrophic interference. Protection requires both input excitation and target demand at the mode’s frequency (Corollary 6.6).
- P10. Nonlinear dwell-time inflation.** In nonlinear recurrent networks, plateau durations are longer than the linear prediction by a factor controlled by the heterogeneity of instantaneous pole-zero separations across operating points (Proposition 9.3).

A Proof Details for the Hessian Structure

We provide the full computation of the Hessian at the cancellation saddle in the general data-dependent setting.

Let $\theta = (W, b, c) \in \mathbb{R}^{n^2+2n}$ be the full parameter vector. At a cancellation saddle where $r_k = \alpha_k \beta_k = 0$ with $\alpha_k = c^\top v_k = 0$ and $\beta_k = u_k^* b \neq 0$ (the case $\beta_k = 0$ is analogous by duality), we work in the modal parameterisation (α_k, β_k) with all other coordinates held fixed.

The loss restricted to (α_k, β_k) near the saddle, for squared error under data distribution $\mathcal{D}$, is (from (24)):

$$\mathcal{L}(\alpha_k, \beta_k) = \mathcal{L}_\perp + |\alpha_k|^2 |\beta_k|^2 S_k^{(x)} - 2 \Re[\alpha_k \beta_k \bar{J}_k] + O(|\alpha_k \beta_k|^2), \quad (51)$$

where the mode energy $S_k^{(x)}$ and cross-energy J_k are defined in Definition 2.3, and $\mathcal{L}_\perp$ is the loss contribution from all other modes.

At $\alpha_k = 0, \beta_k = \beta_k^{(0)}$:

$$\frac{\partial \mathcal{L}}{\partial \alpha_k} = 0, \quad \frac{\partial \mathcal{L}}{\partial \beta_k} = 0, \quad (52)$$

$$\frac{\partial^2 \mathcal{L}}{\partial \alpha_k \partial \bar{\alpha}_k} = |\beta_k^{(0)}|^2 S_k^{(x)} > 0, \quad (53)$$

$$\frac{\partial^2 \mathcal{L}}{\partial \alpha_k \partial \bar{\beta}_k} = -\bar{J}_k, \quad (54)$$

$$\frac{\partial^2 \mathcal{L}}{\partial \beta_k \partial \bar{\beta}_k} = 0. \quad (55)$$

The real Hessian restricted to $(\Re \alpha_k, \Im \alpha_k, \Re \beta_k, \Im \beta_k)$ has eigenvalues (each with multiplicity 2):

$$\mu_\pm = \frac{|\beta_k^{(0)}|^2 S_k^{(x)}}{2} \pm \sqrt{\frac{|\beta_k^{(0)}|^4 (S_k^{(x)})^2}{4} + |J_k|^2}. \quad (56)$$

Since $S_k^{(x)} > 0$ (any nondegenerate input distribution has positive energy in every mode) and $|J_k| > 0$ (hypothesis (c) of Theorem 3.3), we have $\mu_+ > 0$ and $\mu_- < 0$, confirming the saddle.

Note that the Hessian eigenvalues depend on the data distribution through $S_k^{(x)}$ and J_k . The escape rate $\mu_- \propto -E_k^{\text{eff}}$ determines the instability of the saddle: modes with higher effective spectral energy produce more unstable saddles and faster escape.

B Relationship to the Saxe–McClelland–Ganguli Framework

We make explicit the formal correspondence between the feedforward theory of Saxe et al. and the recurrent theory developed here.

Deep Linear Network (Saxe)	Linear Recurrent Network (This work)
End-to-end map $W_L \cdots W_1$	Transfer function $H(z; \tau) = C(zI - W)^{-1}B$
Singular values σ_k	Residues r_k at poles p_k
Rank of $W_L \cdots W_1$	Effective rank $\rho = \#\{k : r_k \neq 0\}$
Saddle: $\sigma_k = 0$	Saddle: $r_k = 0$ (pole-zero cancellation)
Activation: $\sigma_k : 0 \rightarrow >0$	Activation: pole-zero separation
Ordering by target SVD	Ordering by $E_k^{\text{eff}} = E_k^{\text{target}} \cdot S_x$
Dwell time $\sim 1/\sigma_k^{*2}$	Dwell time $\sim 1/E_k^{\text{eff}}$
Singular values $\in \mathbb{R}_{\geq 0}$	Residues $\in \mathbb{C}$
No topological invariant	Winding number $\mathcal{N}(\gamma_k) \in \mathbb{Z}$
No spiral transients	Spiral separation for complex poles
No catastrophic interference mechanism	Non-partner collision
Loss-independent ordering	Input-dependent ordering

The central conceptual advance is the replacement of the singular value (a non-negative real number) with the pole-zero configuration (a pair of points in $\mathbb{C}$). The additional structure of $\mathbb{C}$ —the argument principle, residues, the geometry of collisions in two dimensions—provides a richer and more predictive theory. The data-dependent ordering through E_k^{eff} is a genuinely new prediction: in Saxe’s framework, the singular values of the target determine the ordering independently of the input statistics.